

Conditional entropy

$$H(Y) = -\sum_i p(y_i) \log_2 p(y_i)$$

$$H(Y | X = x_j) = -\sum_i p(y_i | x_j) \log_2 p(y_i | x_j)$$

$$H(Y | X) = \sum_j p(x_j) H(Y | X = x_j)$$

$$= -\sum_j p(x_j) \sum_i p(y_i | x_j) \log_2 p(y_i | x_j)$$

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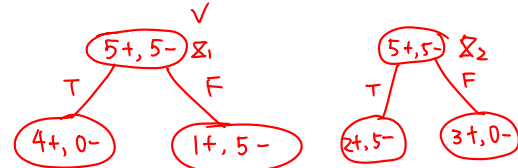
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Example

Attributes		Labels	
X1	X2	Y	Count
T	T	+	2
T	F	+	2
F	T	-	5
F	F	+	1

Which one do we choose

X1 or X2?



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X1	X2	Y	Count
T	T	+	2
T	F	+	2
F	T	-	5
F	F	+	1

$H(Y | X_1 = T) = - \left(\frac{4}{10} \log_2 \frac{4}{10} + \frac{6}{10} \log_2 \frac{6}{10} \right)$
 $\Rightarrow - \left(\frac{4}{10} \log_2 \frac{4}{10} + \frac{6}{10} \log_2 \frac{6}{10} \right)$
 $= 0$

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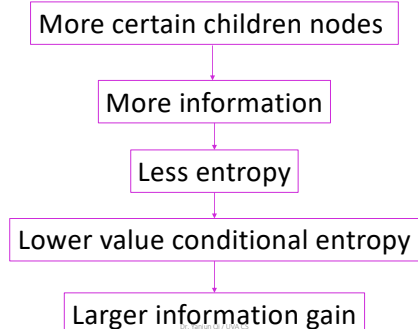
$H(Y | X_1 = T) = \left(\frac{4}{10} \right) \Rightarrow - \left(p(+) \log p(+) + p(-) \log p(-) \right)$
 $= - \left(1 \log 1 + 0 \log 0 \right) = 0$
 $H(Y | X_1 = F) = \left(\frac{1}{5} \right) \Rightarrow - \left(p(+) \log p(+) + p(-) \log p(-) \right)$
 $= - \left(\frac{1}{5} \log \frac{1}{5} + \frac{4}{5} \log \frac{4}{5} \right)$

$H(Y | X_1) = \frac{4}{10} H(Y | X_1 = T) + \frac{6}{10} H(Y | X_1 = F)$

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Intuition of Node Splitting



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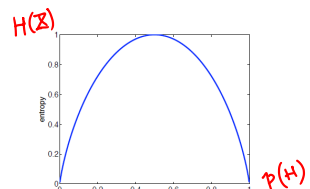
Entropy

- If there are k possible outcomes

$$H(X) \leq \log_2 k$$

- Equality holds when all outcomes are equally likely

- The more the probability distribution that deviates from uniformity, the lower the entropy



$$H(X) = E(I(X)) = \sum_i p(x_i) I(x_i) = -\sum_i p(x_i) \log_2 p(x_i)$$

e.g. for a random binary variable

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